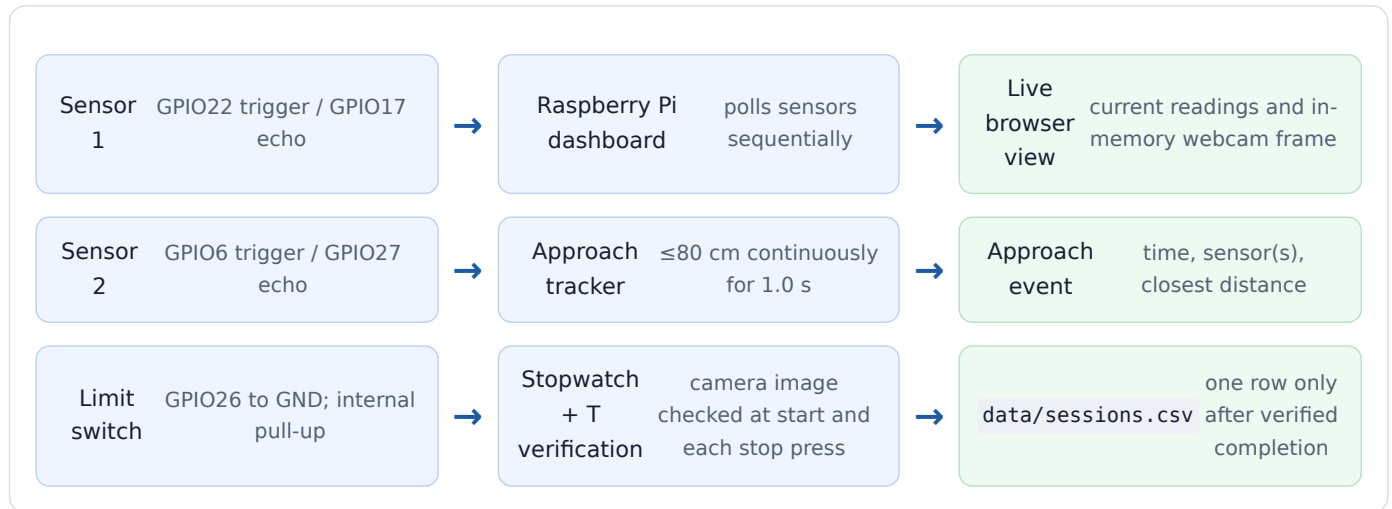


Current system flow

Puzzle Telemetry Rig, Phase 1 — the live Raspberry Pi dashboard, two ultrasonic sensors, limit switch, webcam, completion check, and session CSV.

Current design, rev 4. This document describes the running dashboard flow. Wiring remains controlled by `docs/wiring.md`; runtime behaviour is controlled by `src/tools/live_dashboard.py`.

1. What is connected and what runs



Both HC-SR04 Echo signals pass through their own 1 kΩ + 2 kΩ voltage divider before reaching the Pi. The camera is USB. The e-paper display is currently disconnected.

2. Session flow

1

READY

Dashboard polls both sonars and waits for a switch press. A sustained nearby reading may create a pending approach.

2

START CHECK

First press takes the current camera frame and checks that the puzzle is *not* already complete.

3

TIMING

For an incomplete puzzle, record `start_ts`, attach the pending approach if present, and run the stopwatch. New approaches are paused.

4

STOP CHECK

Every later press checks the live camera frame against the saved solved-T reference. It must pass before the deadline.

5

DONE

Only a passing reference check before the deadline stops the timer and appends the session row. The next press begins a new start check.

Normal route: first press → puzzle is incomplete → timing starts. Later press → T is complete → save the metrics and finish.

Already solved at first press: the system returns to READY. It does not start the timer and does not create a session row.

Incomplete on a stop press: the timer keeps running. No session row is written; the visitor can continue and press again later.

No successful verification within 5 minutes: the rig saves one row with `completed=abandoned`, returns to READY, and does not carry an approach into the next session.

3. Metrics and current data collection

Metric / data	When captured	Current behaviour
<code>session_id</code>	Verified completion	Anonymous sequential identifier.
<code>approach_ts</code>	When one sonar stays at or below 80 cm for 1.0 s	Saved only if that approach is linked to a valid start.
<code>start_ts</code> , <code>stop_ts</code>	Valid start / verified completion	ISO 8601 local timestamps.
<code>approach_to_start_s</code>	Valid start	Time from sustained approach to the first accepted press; blank when no approach was available.
<code>solve_time_s</code>	Verified completion	Elapsed stopwatch time from valid start to successful verification.
<code>min_distance_cm</code>	During recorded approach	Closest valid reading from the detecting sonar(s).
<code>completed</code>	Verified completion	Written as <code>yes</code> by the current dashboard flow.
Shape score, area ratio, detecting sensor(s)	Verified completion	Saved inside <code>notes</code> for diagnosis and threshold tuning.
Current sensor distances and switch events	Live dashboard	Shown live, not written as a per-sample telemetry log.
Video, continuous camera frames, faces	Never	Not stored. The dashboard uses only the current in-memory JPEG frame.
<code>hand_frames_pct</code> , <code>frames_total</code> , <code>verify_photo</code>	Not implemented in dashboard	Columns exist in the schema but are currently blank.

4. Edge cases and safeguards

Situation	Current behaviour
One sensor returns <code>NO ECHO</code>	That reading is ignored. The other sensor can still create an approach event.
Brief pass near the table	Not recorded as an approach unless proximity remains at or below 80 cm for the full 1.0-second hold period.
Approach is not followed by a start	It expires after 60 seconds, so its values will not be attached to a later session.
Person moves away after an approach	After 10 seconds away from the threshold, the stored approach resets.
Switch bounce	GPIO26 uses 50 ms debounce, so ordinary contact bounce should not generate multiple presses.
Start pressed while the T is already solved	Timer remains stopped, no row is written, and the dashboard asks for the puzzle to be reset.
Stop pressed while T is incomplete	Timer continues; no incomplete result is saved as a completed session.
Camera check finishes after the 5-minute deadline	The deadline wins: the session is saved as <code>abandoned</code> , not as a successful solve.
Camera frame unavailable or completion check errors	At start: return to READY without timing. At stop: keep timing and ask for another check.
CSV write fails after a complete T	The dashboard displays an error; the completed session was not saved and should be investigated before the next visitor.
Both sonars hear each other	The code triggers and reads them in sequence, rather than pinging both at once, reducing ultrasonic cross-talk.

5. Current limitations to keep in mind

Completion accuracy

The completion decision is a fixed-camera HSV colour, contour-shape, and area comparison. It should be validated with labelled complete and incomplete examples under the final lighting before relying on it for formal measurement.

Configured but not yet active

`result_hold_s` is reserved for the future e-paper display. The dashboard keeps a result visible until the next switch press; it does not automatically clear after 20 seconds.

6. Configuration that controls the flow

`src/config.yaml` → `session` is the source of these values: `approach_cm: 80`, `approach_hold_s: 1.0`, `approach_reset_s: 10`, `near_timeout_s: 60`, and `safety_timeout_s: 300`. The T-completion HSV threshold, contour shape threshold, and area-ratio threshold are in the same file under `completion`.

Current wiring source of truth: `docs/wiring.md`. Current runtime flow source: `src/tools/live_dashboard.py`.